

5.15

Create a function and then use central difference to calculate the derivative of the function in the range of $-2 < x < 2$.

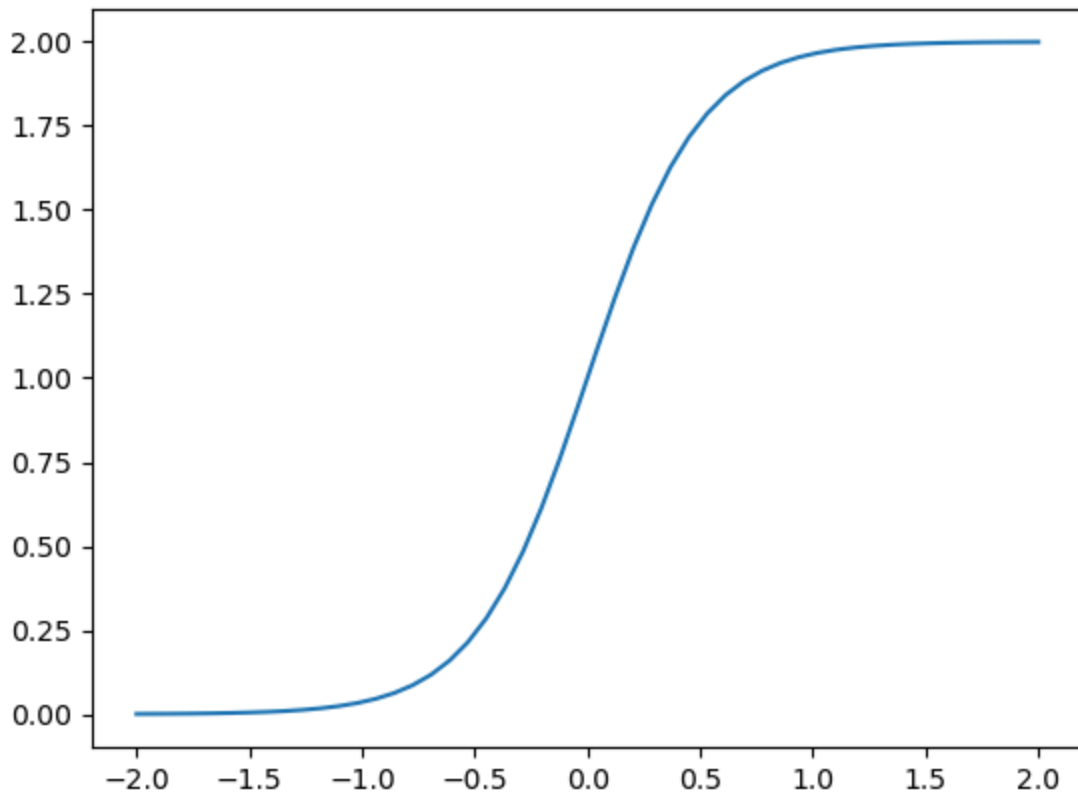
```
In [1]: import numpy as np
import matplotlib.pyplot as plt
```

```
In [6]: def f(x):
        return(1+np.tanh(2*x))

x = np.linspace(-2,2)

# plot
fig, ax = plt.subplots()
ax.plot(x, f(x))
```

```
Out[6]: [<matplotlib.lines.Line2D at 0x784daf641100>]
```

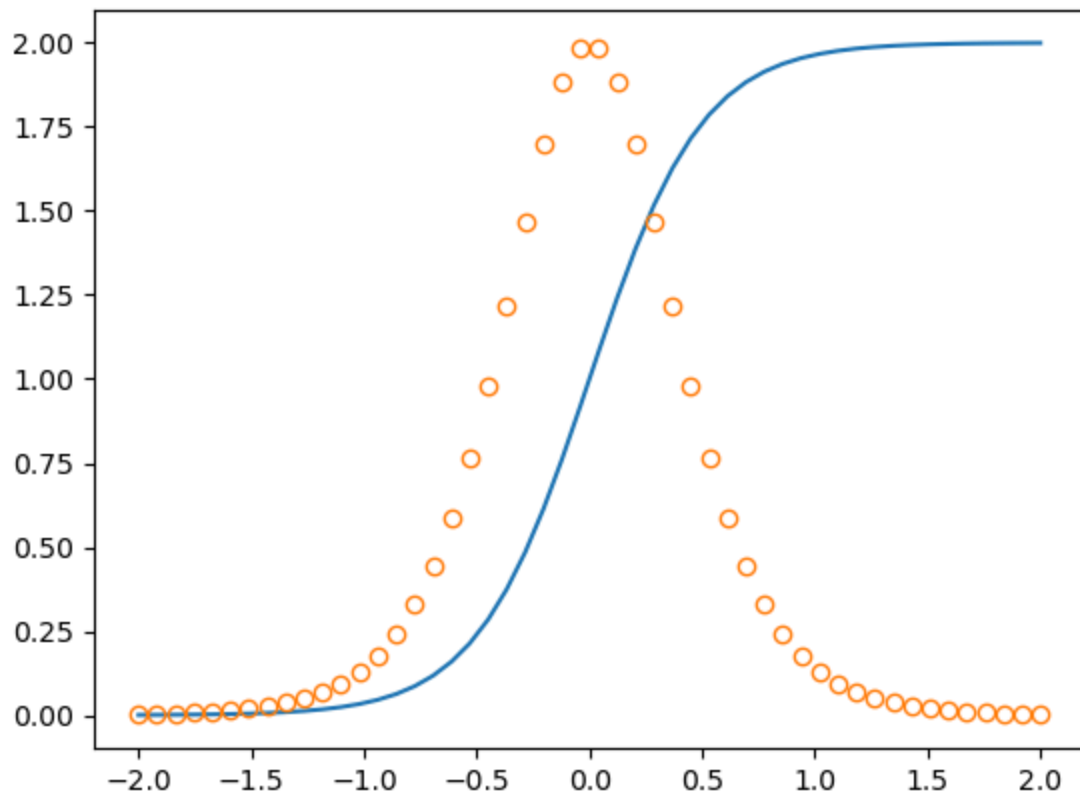


```
In [8]: def mid(function, x_value, stepSize):
        return (function(x_value+stepSize)-function(x_value-stepSize))/(2*stepSi

derivative = []
for i in x:
    f_prime = mid(f, i, 0.001)
    derivative.append(f_prime)
```

```
fig1, ax1 = plt.subplots()
ax1.plot(x, f(x))
ax1.plot(x, derivative, 'o', mfc='none')
ax1.legend([r'f(x)', r'df/dx'])
```

Out[8]: <matplotlib.legend.Legend at 0x784daf52bf20>



In []: